



BASE II Clearing Overview

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Visa Supplemental Requirements

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Chapter 1

BASE II Clearing System Overview

BASE II Clearing System is the international electronic clearing system supporting transaction-based payment exchange between Visa clients.

Clearing is the process by which Visa collects data about a transaction from the source, validates the information and calculates fees, values the transaction (calculates the base value of the transaction as well as various fees and charges), and then delivers it to the destination. The destination uses this information to post the transaction to the cardholder's account or reconcile a merchant's settlement position.

BASE II Clearing System is one of Visa's core processing functions, ensuring that merchants get paid and enabling clients to manage the accounts of cardholders. From a business standpoint, the clearing system is complex due to the number of regions it covers and the need to adapt the system to keep up with specialized arrangements by clients, unique banking regulations, legal agreements, and changing political landscapes.

BASE II's strengths include:

- **Brand-Agnostic Processing** – It supports all cards (such as Visa, Mastercard, and American Express) and several non-card payment services. BASE II will edit, assess fees, and create the necessary settlement information without considering what brand name the transaction occurred under. It also performs processing based on unique rule sets requested by clients.
- **Continuous Clearing and Settlement Processes** – BASE II is a transaction engine that is continually running: collecting files, stripping out records, creating trail balances and settlement positions. At any one time, it has upwards of 200 collection files open and being processed.
- **Extraordinary Power and Capacity** – BASE II's role as a core processing component means it must run with 100% reliability while it processes an average of about 100 million transactions per day (2006 statistic). In addition to client-to-client interfaces, BASE II also supports exchange/transport of data for many other Visa services, in total managing an average of 1.5 billion records daily (2006 statistic).

BASE II VisaNet Interchange Overview

BASE II Clearing System collects and delivers Visa transaction data, often referred to as an interchange, through the Visa transaction processing and communications network known as VisaNet.

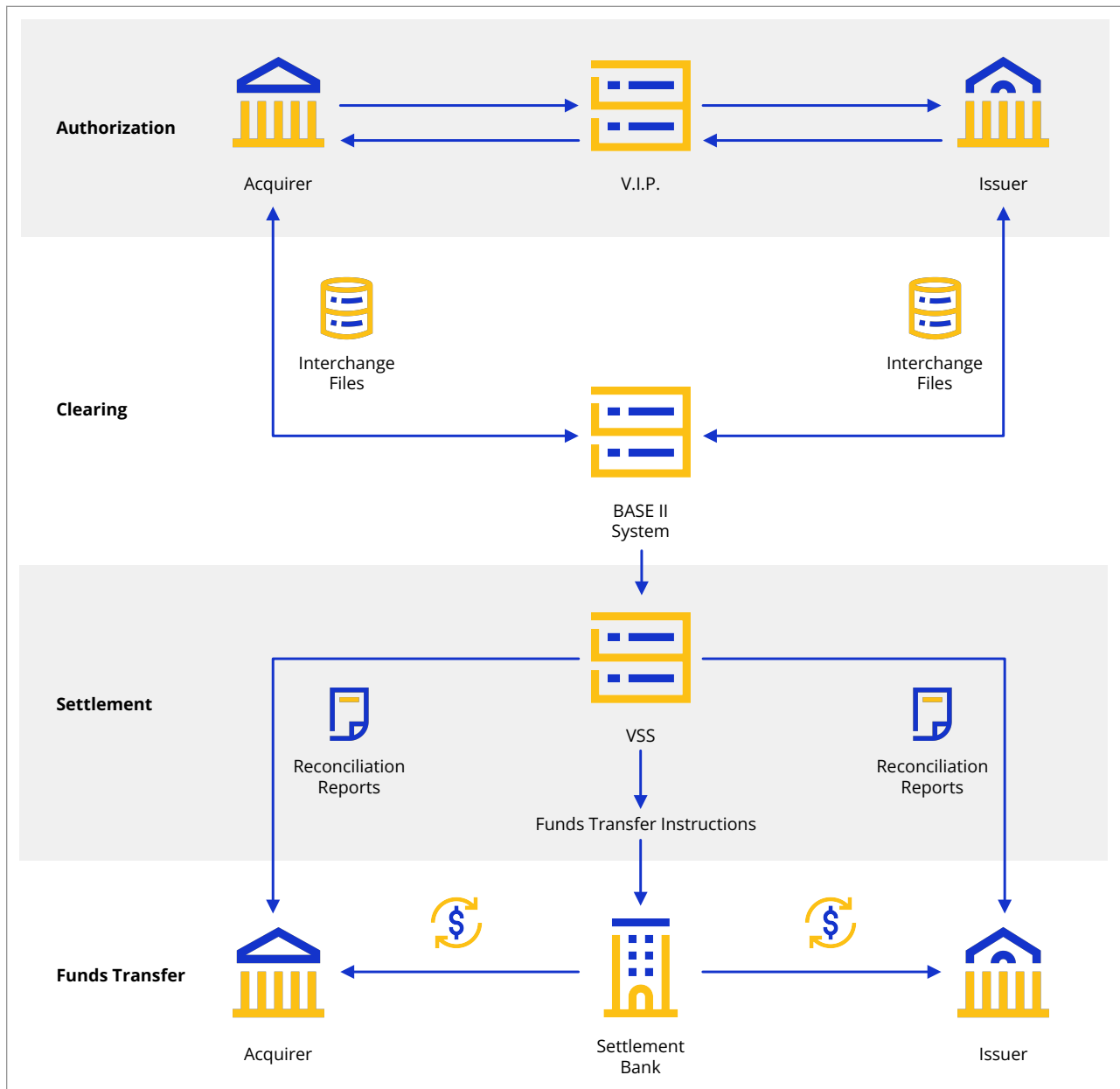
VisaNet is the processing platform that connects Visa and its 21,000 worldwide client banks and financial institutions so that they can authorize and settle electronic payments quickly and securely. The VisaNet network comprises hardware and software, including BASE II, which perform transaction processing and are attached to communications facilities that connect with clients' systems and other networks. The network has a low-risk, redundant infrastructure (for example, backup generators, duplicate equipment, multiple routes for a transaction to get to its destination) that eliminates single points of failure and enables it to be available 24 hours a day.

Depending on the region, VisaNet supports a wide variety of transactions, including: purchase; cash and bill payment transactions made with any card; cardholder transfer; recurring payment transactions; and ATM transactions for other networks, such as Plus. These transactions are processed through VisaNet's authorization (V.I.P.), clearing (BASE II), and settlement services.

The flow of the services is:

1. **Authorization** – The process by which an issuer approves or declines a cardholder's transaction before a purchase is finalized or cash is disbursed.
2. **Clearing** – The process by which Visa collects transaction data from the acquirer, validates the transaction, enforces risk services, calculates fees and charges, and delivers the validated information to the issuer for posting to the cardholder's account.
3. **Settlement** – The process of accumulating advices from clearing, determining each client's net settlement position (in other words, who owes what to whom), and initiating the exchange of funds. Providing settlement information to clients and processors in standardized reports.

VisaNet Authorization, Clearing, and Settlement Flow



The term VisaNet applies to all components of the network including hardware, software, communications facilities, and the systems that perform transaction processing and other services.

The main components of VisaNet are:

- VisaNet Interchange Centers (VICs)
- VisaNet Access Points (VAPs or Extended Access Servers) and other network connections
- Telecommunications Network

BASE II VICs

BASE II Clearing System uses VisaNet Interchange Centers (VICs), also known as VisaNet Data Centers, to maintain and operate the mainframe computers and the Inovant software applications that perform all VisaNet transaction processing

VICs are large-scale computer operations centers that maintain and operate the mainframe computers and the Inovant software applications that perform all VisaNet transaction processing. Each VIC serves as the control point and housing for the disk storage, printing, and telecommunications equipment necessary to support VisaNet. There are four VICs: two in the United States, one in the United Kingdom, and one in Singapore. Each VIC is linked to all the other VICs, enabling each one to serve as a backup in the event of system interruptions.

BASE II VisaNet Connections

Clients connect to VisaNet both for transaction processing and for file delivery. A client can connect to any combination of VisaNet systems, based on their system requirements and their processing choices.

Clients have a number of connectivity choices to link to VisaNet.

- **VisaNet Access Point (VAP)** – PC-based access device that connects the host computer in the client's processing center to VisaNet.
- **Extended Access Server (EAS)** – Servers located at endpoints (mostly clients and processors), or in VICs, and are used to provide a secure, single point of connectivity to VICs, the Direct Exchange network, and/or VisaNet Distributed Processing Solution (VDPS). EAS performs authorization routing, file staging and delivery services, and provide secure connectivity to VisaNet
- **Direct Exchange (DEX)** – Internet Protocol (IP) – based technologies that provide Visa endpoints with a single point of connectivity for file delivery, message routing, and integrated, web-based payment and information solution sets, delivered over multiple channels.

BASE II VisaNet Communications Network

BASE II Clearing System uses a private IP VisaNet communications network to provide a worldwide data communications system to client processing centers and VICs.

Client processing centers and the VICs are connected to each other through a private IP VisaNet communications network. This worldwide data communications system controlled by Visa links clients and the VICs for transaction traffic. It is also used to perform remote diagnostics and provide maintenance and backup services.

Inovant has contracted with both AT&T and Verizon to provide telecommunications between the various parts of VisaNet so if one provider experiences a service outage, the system automatically shifts to the other provider.

BASE II Member Processing Centers

BASE II Clearing System uses VisaNet to provide authorization, clearing and settlement services, message control, stand-in authorization, and backup services to member processing centers.

While not technically part of VisaNet, member processing centers are a key component of transaction processing. A processor is an endpoint that has back-office systems it runs on its own behalf or on behalf of financial institutions. (An endpoint is a logical processor, and also the physical location of that processor.)

Processing centers (also known as VisaNet endpoints) are the physical data processing facilities owned by members or by third-party processors and linked to VICs for authorization, clearing and settlement services, message control, stand-in authorization, and backup services. The processing center houses card processing systems that support merchant and business locations and cardholder billing.

Each processing center has a physical connection to VisaNet, facilitated by a VAP, Extended Access Server, or other connectivity device, over which it sends the data to be processed (outgoing interchange) and receives data back from Visa (incoming interchange). To ensure system redundancy and security, all VisaNet endpoints connect to two VICs: one serves as the primary VIC, processing all or most transactions, while the other is the secondary (or backup) VIC.

BASE II VisaNet Software Components

BASE II Clearing System uses software components to provide data exchange.

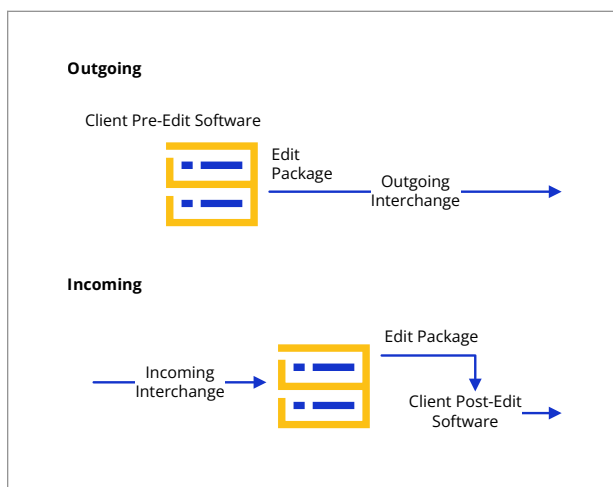
The exchange of BASE II data via VisaNet involves this software:

- Processing center software
- Connectivity software

BASE II Processing Center Software

The primary software components used by the client's center for BASE II processing are the pre-edit programs, the Edit Package, and the post-edit programs.

BASE II Processing Center Software



- **Pre-Edit Software** – Written by processing center personnel or by an outside vendor, pre-edit software is generally part of a merchant processing system or cardholder system. Although pre-edit software differs from center to center, this software has a common purpose, which is to place transaction data into a format that is acceptable to the Edit Package.
- **Edit Package** – Provides a bridge between endpoint systems and VisaNet so that endpoints can conform to Visa requirements with less impact to their own systems. Supplied by Visa, Edit Package is a series of programs that are operated daily by processing centers to prepare their interchange data files and to verify the integrity of the files they receive from Visa. Visa supplies Edit Package software that is compatible with various computer systems made by major manufacturers.

The Edit Package is the interface between the endpoint's processing center's internal payment processing systems and the file transfer process used to interchange files with BASE II. The Edit Package prepares outgoing interchange for transmission to the VIC by validating the information and ensuring that it is free of obvious error. Edit Package programs consist of four modules:

- **Edit Package Initializer (EDI)** – Sets up routing tables and control data needed by the editing programs.
- **Edit Package Processor (EDP)** – Performs editing functions, converts transaction formats, and produces reports.
- **Summary Report Writer (EDS)** – Produces summary reports.
- **Report Generator (ERG)** – Produces optional reports during the incoming processing cycle.

Edit Package also receives incoming interchange from the VIC, prepares the interchange for final processing by the processing center's system, and produces audit trail and error reports.

- **Post-Edit Software Edit Package** – Written by processing center personnel or by an outside vendor, post-edit software reformats and processes incoming interchange received from the VIC. This software prepares interchange so that it can be processed by cardholder and merchant posting programs and used for other processing center purposes, such as fraud analysis.

BASE II Visa Markup Language

Visa Markup Language (VML) is an alternative clearing file format designed to operate with the BASE II Clearing System.

Patterned on XML, VML contains the same fields and data used in fixed-length records but they are encoded within nested VML tags, which enable variable-length records. With VML, the concept of a fixed Transaction Component Record (TCR) no longer applies.

Currently, VML can only be used by endpoints that are not using the Edit Package option.

Note: VML is no longer being supported.

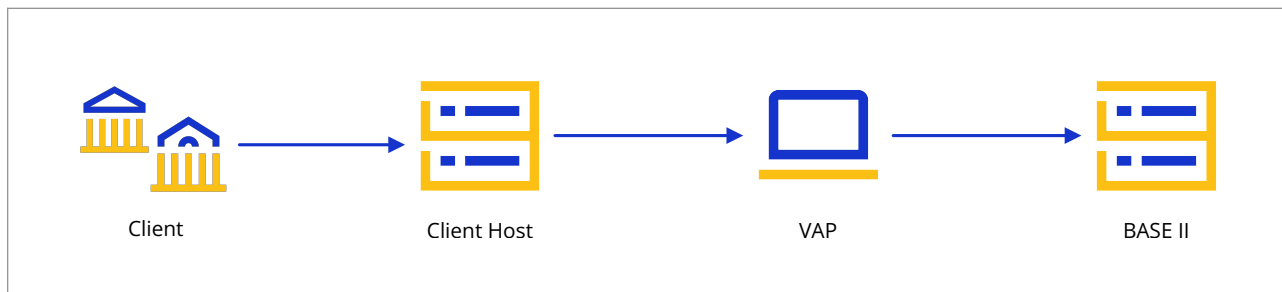
BASE II Connectivity Software

BASE II Clearing System provides a number of connectivity options to link to VisaNet (VAP, DEX, Extended Access). Each software option uses its own unique software to manage the transfer of files.

BASE II VAP Software

VAP is a PC with Visa-supplied software that supports and monitors the transfer of BASE II files to and from the VIC. As part of interchange file management, the VAP software detects transmission errors, performs automated error recovery, and records BASE II file information and transmission-related messages. Its proprietary communication protocol and compression algorithm with CAS Transport Services (TSA) ensures speed and security of processing, and seamless error recovery without client intervention.

VAP to BASE II Flow



BASE II Extended Access Server

The Extended Access Server works similarly to the VAP but has more security options built in. It uses third-party software to provide services such as Tivoli (TEC) feed, web server, database, SSL encryption/decryption, and an audit trail. The third-party software comes pre-installed and configured on an EAS and includes an intuitive, color-coded graphical user interface that provides a consolidated view of the client's messaging and file service activities. This component also employs a proprietary communication protocol and compression algorithm with CAS Transport Services (TSA), ensuring speed and security of processing as well as seamless error recovery without client intervention.

BASE II Clearing System Documentation

This book is part of the set of BASE II Clearing documents. *BASE II Clearing System Overview* is designed to be a companion to *BASE II Clearing System Services*, which describes the various BASE II-related services offered. Refer to the BASE II Clearing documents below for more information on BASE II.

BASE II Clearing Data Codes

BASE II Clearing Interchange Formats, TC 01 to TC 49

BASE II Clearing Interchange Formats, TC 50 to TC 92

BASE II Clearing Edit Package Operations Guide Release 4

BASE II Clearing Edit Package Messages Reference Guide Release 4

BASE II Clearing PC Edit Package for Windows User Guide Release 4

BASE II Clearing Edit Package Reports Reference Guide Release 4

Note: VML is no longer being supported.

Chapter 2

BASE II Clearing System Workflow

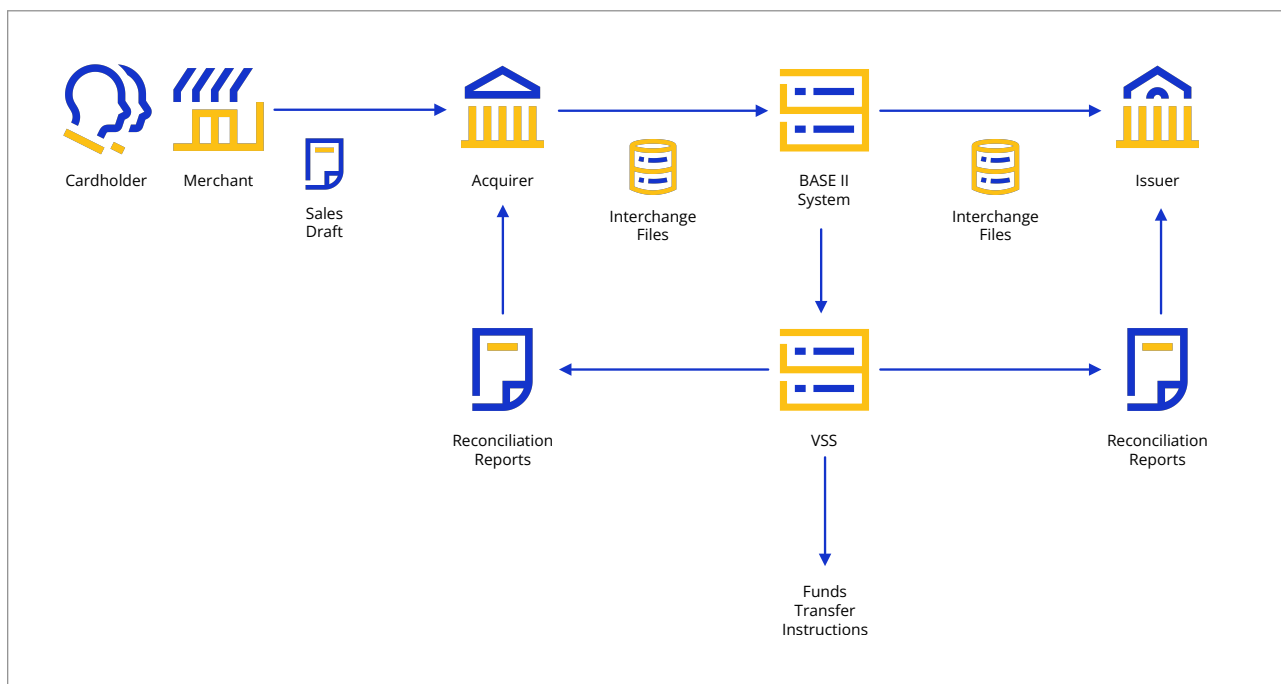
BASE II Clearing System receives and processes interchange files from clients at intervals specified by the clients and provides the settlement information at the appropriate time to receiving institutions, clients, and processors.

The workflow between merchants, clients, and Visa can be summarized as follows:

- **Merchant**—Transmits the electronic sales drafts to their acquiring institution or processor.
- **Acquiring Institution or Processor**—Edits, formats, and sends the merchant's transactions electronically as interchange files to a VisaNet Data Center.

- **VisaNet Interchange Center**—Clears the acquiring institution or processor's interchange files by:
 - Collecting, editing, and validating the interchange files, transaction by transaction.
 - Performing any necessary currency conversion.
 - Calculating interchange fees and Visa's daily transaction charges.
 - Creating settlement records for VisaNet Settlement Service, (VSS).
 - Creating and routing delivery interchange files to the appropriate issuing institutions and processors
- **Issuing Institution or Processor**—Reviews the clearing transactions and, if appropriate, removes the hold from the cardholder's funds and posts the amount of the transaction to the cardholder's account.

BASE II Clearing Workflow



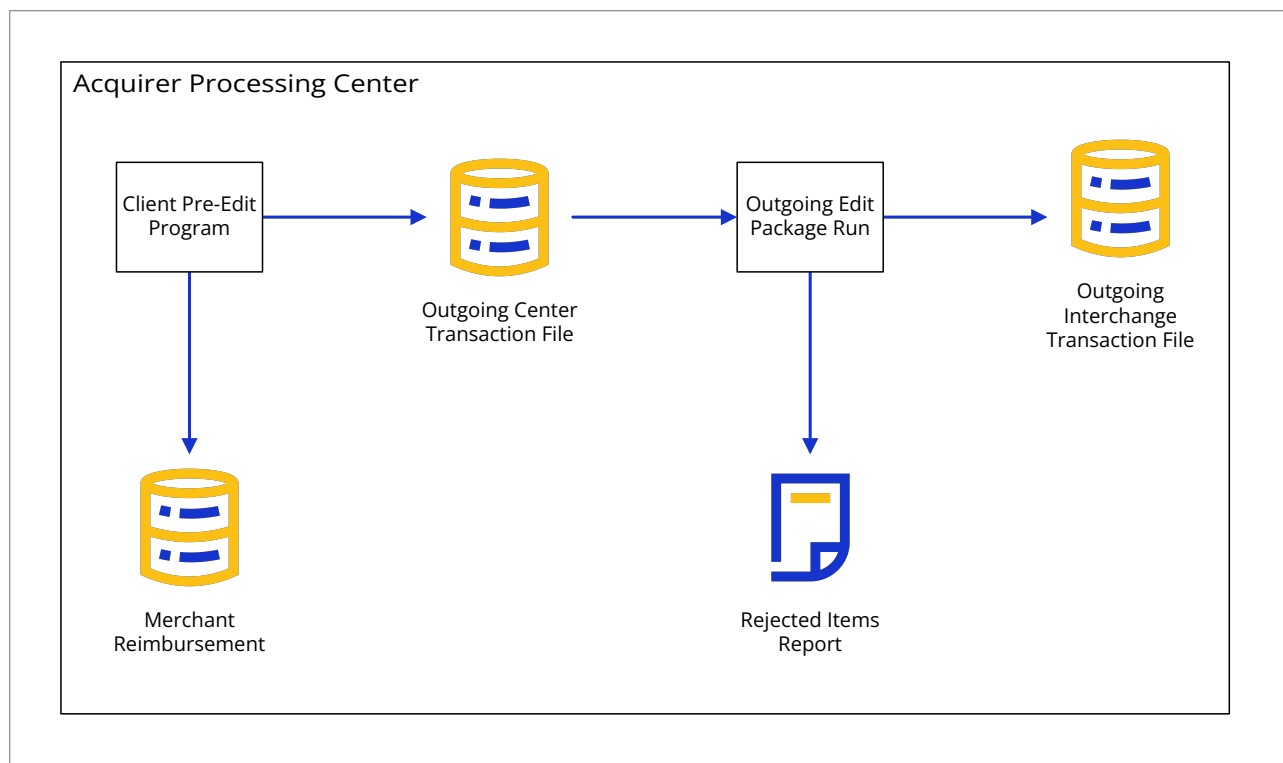
BASE II Transaction Reconciliation

At a client option, Visa can provide transaction-level reconciliation through Clearing and Settlement Advices. These advices are created, one advice per transaction.

BASE II Interchange Transaction File

The acquirer or its processor converts the merchant's transaction data into Visa's format. Some clients and processors run the data through Visa's Edit Package application, while others have created their own editing applications. The edit application rejects items not suitable for collecting into BASE II.

BASE II Interchange Flow



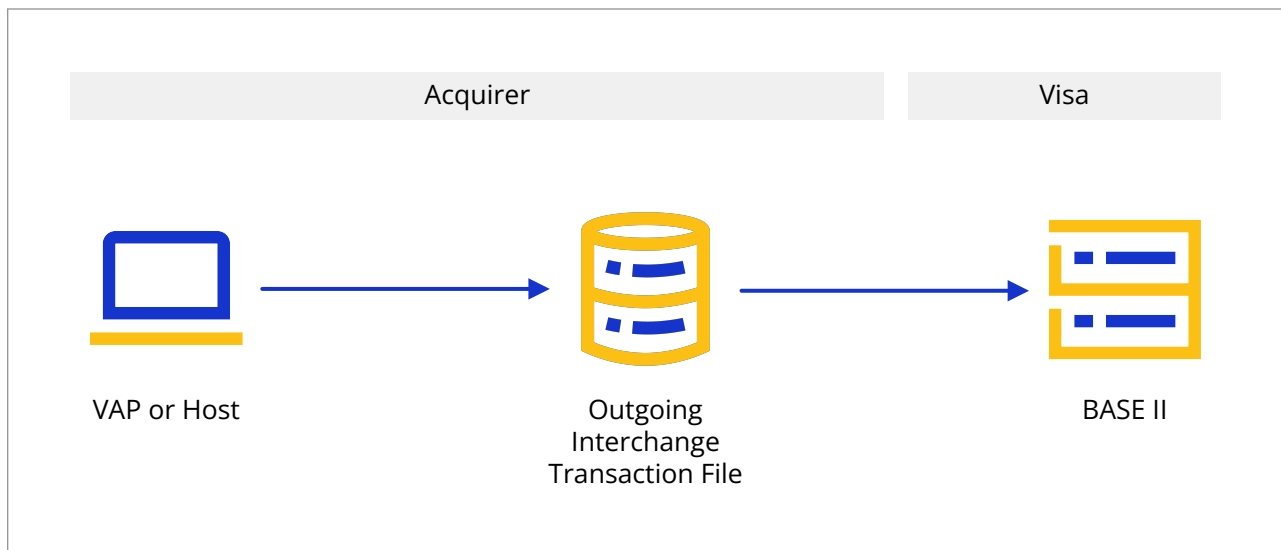
VisaNet Interchange Centers (VIC)

Throughout the day, VisaNet Interchange Centers (VIC) collect and receive interchange files containing transactions from a number of types of connections:

- **VisaNet Access Point (VAP)**—The Application Platform Facility (APF) dials out to the client/processor's access point and collects the files per daily schedule, as specified by the client.
- **Extended Access Server (EAS)**—Extended Access performs similarly to VAP, but includes more advanced security functionality.
- **Visa Direct Exchange (DEX) network**—The client/processor initiates the exchange, sending the files to Visa.
- **VisaNet Integrated Payment (V.I.P.) System**—Generates a record for full financial transactions to the clearing and settlement system.

Collection can be performed at any time specified by the client. BASE II polls the clients/processors to see if they have anything to collect at the specified time.

BASE II Outgoing Interchange from Acquirer



BASE II Interchange Transaction File Processing

BASE II Clearing System receives the Interchange Transaction File and processes the transactions.

When an interchange file reaches BASE II, the records are stripped out independently and processed through the system; by transaction, not by file or batch. Each transaction passes through the clearing system in a matter of milliseconds.

- **Validates Source**—BASE II ensures that it recognizes the parties to the transaction and determines that they have the right to submit interchange.
- **Determines Jurisdiction**—BASE II determines where the transaction took place (domestic, regional, interregional) to establish what rules the transaction falls under and what fees are appropriate.
- **Edits the Transaction**—The data is validated to ensure its integrity (for example, is the card number correct, is the country code valid, etc.)
- **Determines Settlement Service**—The clearing system determines the settlement service to use (international and national, area, and product settlement services), the currency used, and the timing of the settlement. Clients can request a particular settlement service, or allow the system to choose the most appropriate settlement service.
- **Values the Transaction, including Currency Conversion**—The system converts the transaction amount into the possible currencies involved in the transaction and settlement.
- **Calculates the Transaction's Fees and Charges**—Interchange fees are what one client pays another: for purchase transactions, the acquirer pays the issuer, and for ATM/Cash transactions, the issuer pays the acquirer. Interchange fees vary by jurisdiction based on regulations, product, settlement service, network, timeliness of clearing, etc. Daily transaction charges are calculated by Clearing.

VSS Transactions

BASE II Clearing System provides the information that VisaNet Settlement System (VSS) needs for settlement.

BASE II produces these records in the process of clearing a transaction:

- Settlement records for VSS.
- SMS reporting records for transactions that involve a full-service client.

VSS Processing

Visa Settlement System (VSS) uses the individual settlement records to calculate one net position for each funds transfer endpoint in the currency the endpoint has requested.

For example, Globalbank has two endpoints with different currency requirements: Globalbank Europe wants its position in euros, while Globalbank US wants its position in dollars.

VSS produces:

- Reconciliation reports for the issuer and acquirer, summarizing the activity underlying the net settlement amounts.

Note:

Reconciliation reports are also sent to “processing” institutions, which some issuers and acquirers designate to handle settlement for them.

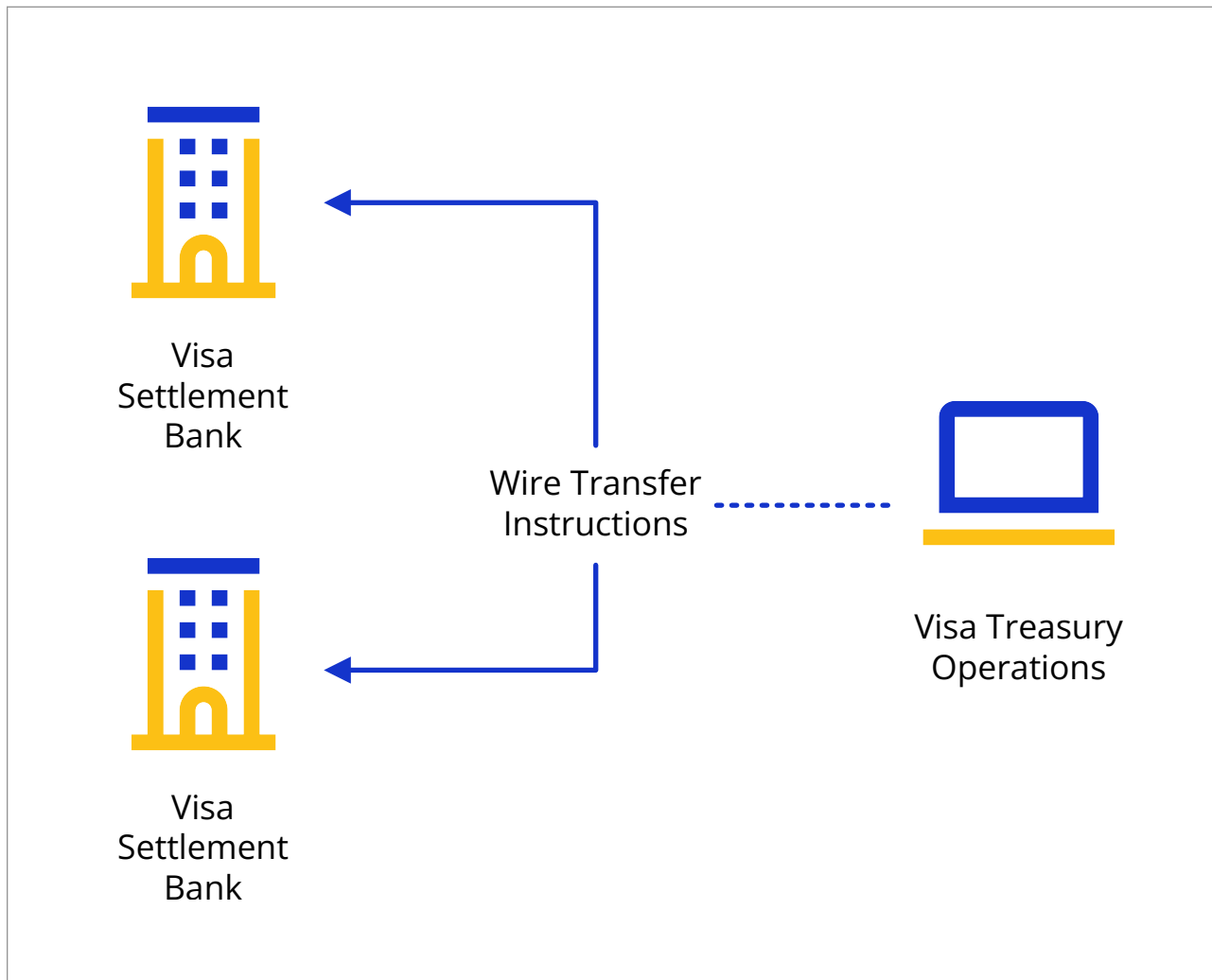
BASE II Visa Treasury Management Department

The Visa Treasury Management Department uses the net position amounts to settle funds for the international settlement service with clients.

Treasury sends funds wire instructions to:

- **Visa settlement banks**—Banks receive settlement wires with funds transfer instructions.
- **Clients that owe money for transactions**—Clients receive requests for funds.
- **Clients that are owed money**—Clients are informed how much they are owed.

Visa Treasury Wires Transfer Instruction to Settlement Banks



Note: Visa supports additional settlement banks depending on the settlement service and financial arrangements

The Visa settlement bank initiates and administers the actual collection of the owed funds:

1. The Visa settlement bank sends the owing client (typically the issuer) a wire, instructing them to pay the net settlement amount:
 - US clients receive their initial instructions via a Federal Reserve wire.
 - Clients outside the US receive a Global wire (aka Swift).
2. The client asks that their bank to wire a specified amount to Visa's settlement account.
3. Visa's Treasury Management Department instructs the settlement banks to pay out the settlement amounts to the clients that are owed money.

Throughout the day, Visa Treasury performs reconciliation and matching the funds requested to the payments. VisaNet Settlement Service (VSS) posts regional fees and charges to the appropriate General Ledger accounts. When wire instructions are sent out, the "value date" on

the wire is the date the client is supposed to pay (1 day for countries that settle using the US dollar, 2 days for countries that settle using other currencies). When a client does not pay on time, Visa will assess late payment fees.

Clearing Settlement Windows

Visa settles with clients per settlement service. Each settlement service has one or more specified times when settlement occurs, referred to as a settlement window. Visa also offers intra-day settlement services for specific conditions.

International Settlement is initiated daily at approximately 03:00 a.m. Pacific time (PT) (10:00 GMT April through October; 11:00 GMT October through April). All client files that have not completed transmission at the time the International Settlement window initiates at approximately 03:00 a.m. PT will be partially processed for inclusion in settlement that day, with the remainder of the file being processed for inclusion in settlement the following calendar day. Clients of other settlement services can designate the time at which these settlement services should close prior to international settlement.

Clearing Settlement Window Closures

This table contains the settlement window in which each settlement service participates, listed by settlement service.

Settlement Window Closures

Settlement Service ID	Settlement Service Name	Time of Closure (GMT)
1	INTERNATIONAL SETTLEMENT	10:00/11:00 ¹
3	SPECIAL SERVICE 3	07:00/08:00 ²
4	SPECIAL SERVICE 4	06:00/07:00 ²
5	SPECIAL SERVICE 5	06:00/07:00 ²
10	AUSTRALIA NATIONAL NET	15:00
11	BARBADOS NATIONAL NET	10:00/11:00 ¹
12	BAHAMAS NATIONAL NET	10:00/11:00 ¹
13	BOLIVIA NATIONAL NET	10:00/11:00 ¹
14	BRAZIL NATIONAL NET ³	10:00/11:00 ¹
15	CANADA NATIONAL NET	10:00/11:00 ¹
16	COSTA RICA NATIONAL NET	10:00/11:00 ¹
17	CZECH REP NATIONAL NET	10:00/11:00 ¹
18	DOMINICAN R NATIONAL NET	10:00/11:00 ¹
19	ECUADOR NATIONAL NET	10:00/11:00 ¹

Settlement Window Closures

Settlement Service ID	Settlement Service Name	Time of Closure (GMT)
20	U.K. NATIONAL NET	06:00/07:00 ²
21	GREECE NATIONAL NET ³	10:00/11:00 ¹
22	GUATEMALA NATIONAL NET	10:00/11:00 ¹
23	HUNGARY NATIONAL NET	10:00/11:00 ¹
24	INDONESIA NATIONAL NET	10:00/11:00 ¹
25	INDIA AREA NET	03:30
26	JAMAICA NATIONAL NET	10:00/11:00 ¹
27	JAPAN NATIONAL NET	02:00
28	SOUTH KOREA NATIONAL NET ³	02:00
29	MEXICO NATIONAL NET	09:00 GMT
30	MALAYSIA NATIONAL NET	10:00/11:00 ¹
31	NEW ZEALAND NATIONAL NET	02:00
32	PERU NATIONAL NET	10:00/11:00 ¹
33	PARAGUAY NATIONAL NET	10:00/11:00 ¹
34	SAUDI AR NATIONAL NET	6:00/7:00 ¹
35	SWEDEN NATIONAL NET	07:00/08:00 ²
36	THAILAND NATIONAL NET	23:00
37	TRINIDAD NATIONAL NET	10:00/11:00 ¹
38	VENEZUELA NATIONAL NET	10:00/11:00 ¹
39	UAE NATIONAL NET	02:00
40	OECS AREA NET	10:00/11:00 ¹
41	KENYA NATIONAL NET	02:00
42	MAURITIUS NATIONAL NET	10:00/11:00 ¹
43	SEYCHELLES NATIONAL NET	10:00/11:00 ¹
44	HONG KONG NATIONAL NET	21:00
45	SINGAPORE NATIONAL NET	23:00
46	ROMANIA NATIONAL NET	10:00/11:00 ¹
49	RUSSIA NATIONAL NET	03:30
50	EGYPT NATIONAL NET	03:30
51	SRI LANKA NATIONAL NET	23:00 GMT ¹
52	POLAND NATIONAL NET	07:00 GMT/06:00 GMT
53	HONG KONG AND CHINA MAINLAND MNSS ³	10:00/11:00 ¹

Settlement Window Closures

Settlement Service ID	Settlement Service Name	Time of Closure (GMT)
54	COLOMBIA NNSS	06:00/07:00 ²
55	EL SALVADOR NATIONAL NET	10:00/11:00 ¹
56	HONDURAS NATIONAL NET	10:00/11:00 ¹
57	PAKISTAN NNSS	03:30
58	CHILE NATIONAL NET	10:00/11:00 ¹
59	ZIMBABWE NATIONAL NET ³	10:00/11:00 ¹
64	PHILIPPINES NATIONAL NET	10:00/11:00 ¹
65	BELIZE NATIONAL NET	10:00/11:00 ¹
67	KAZAKHSTAN NATIONAL NET	02:00
69	CROATIA NATIONAL NET	10:00/11:00 ¹
70	NICARAGUA NATIONAL NET	10:00/11:00 ¹
71	CAYMAN IS NATIONAL NET	10:00/11:00 ¹
72	MACEDONIA NATIONAL NET	23:00
73	BRAZIL CASH DISB NATL NET	06:00/08:00 ⁴
74	MACAU NATIONAL NET ³	10:00/11:00 ¹
76	SERBIA NNSS	06:00/07:00 ²
77	BOSNIA & HERZEGOVINA NNSS	06:00 GMT/07:00 GMT ¹
78	BULGARIA NNSS ³	10:00/11:00 ¹
79	BOTSWANA NNSS	03:30 GMT
80	EURO AREA NET	06:00/07:00 ²
81	UKRAINE NNSS	03:30
82	BELARUS NNSS	10:00/11:00 ¹
84	BANGLADESH NATIONAL NET	23:00 ¹
85	SOUTH AFRICA NNSS	06:00/07:00 ²
86	GEORGIA NNSS	06:00/07:00 ²
87	ALBANIA NNSS	06:00 GMT/07:00 GMT ¹
88	MOZAMBIQUE NNSS	06:00/07:00 ¹
89	GHANA NNSS	10:00/11:00 ¹
90	ZAMBIA NNSS	3:30 ¹
91	TANZANIA NNSS	10:00/11:00 ¹
92	NIGERIA NNSS	02:00
93	VIETNAM NATIONAL NET	02:00

Settlement Window Closures

Settlement Service ID	Settlement Service Name	Time of Closure (GMT)
94	UGANDA NNSS	10:00/11:00 ¹
95	UZBEKISTAN NNSS	03:30 GMT ¹
102	ARMENIA NNSS ³	10:00/11:00 ¹
103	LEBANON USD NNSS ³	06:00/07:00 ²
104	LEBANON LBP NNSS	06:00/07:00 ²
106	MONTENEGRO NNSS ³	10:00/11:00 ¹
107	AZERBAIJAN NNSS	03:30
109	MOLDOVA NNSS	03:30
110	MALAWI NNSS	10:00/11:00 ¹
112	QATAR NNSS	02:00
116	KYRGYZSTAN NNSS	02:00
119	TAJIKISTAN NNSS	10:00/11:00 ¹
120	ANGOLA NNSS ³	03:30
122	TURKEY NNSS ³	06:00/07:00 ²
124	BRAZIL REDECARD NNSS	10:00/11:00 ¹
125	BRAZIL CIP NNSS	06:00/08:00 ⁴
126	RWANDA NNSS	10:00/11:00 ¹
127	BRAZIL CDP NNSS ³	10:00/11:00 ¹
128	GUYANA NNSS	10:00/11:00 ¹
129	MOROCCO NNSS ³	06:00/07:00 ²
131	ICELANDIC NATIONAL NET	07:00
132	MYANMAR NNSS	23:00
133	HAITI NNSS	10:00/11:00 ¹
134	COLOMBIA VCSS NNSS	06:00/07:00 ²
135	DUTCH CARIBBEAN ANSS	10:00/11:00 ¹
136	NEPAL NNSS	03:30
137	ARGENTINA NNSS	10:00/11:00 ¹
138	CENTRAL AFRICAN AREANET ³	06:00/07:00 ²
139	CAMBODIA NNSS	02:00
140	DENMARK NNSS	07:00/08:00 ²
141	MALDIVES NNSS	02:00
142	ARUBA NNSS ³	10:00/11:00 ¹

Settlement Window Closures

Settlement Service ID	Settlement Service Name	Time of Closure (GMT)
143	NORWAY NNSS	07:00/08:00 ²
144	URUGUAY UYU NNSS	10:00/11:00 ¹
145	URUGUAY USD NNSS	10:00/11:00 ¹
146	CAMBODIA (GOV) NNSS	02:00
147	PAPUA NEW GUINEA NNSS	19:00
148	JORDAN NNSS	03:30
149	FIJI NNSS	19:00
150	IRAQ NNSS ³	03:30
151	ARGENTINA NNSS USD	10:00/11:00
152	BOLIVIA NNSS	10:00/11:00

- 1 Adjusts with Pacific daylight savings time.
- 2 Adjusts with Central Europe daylight savings time.
- 3 Inactive as of this publication's effective date.
- 4 Adjusts with Brazil daylight savings time.

Client files containing transactions for a given window not received by Visa at the time of Settlement Window closure will not be processed until the next Central Processing Date (CPD).

BASE II National Net Settlement

Client banks may opt to participate in National Net Settlement Services (NNSS). VisaNet Settlement Service (VSS) determines their positions and transmits their settlement information at the time they have specified through their settlement agent. Each NNSS/Area Net Settlement Service (ANSS), therefore, has a settlement window time.

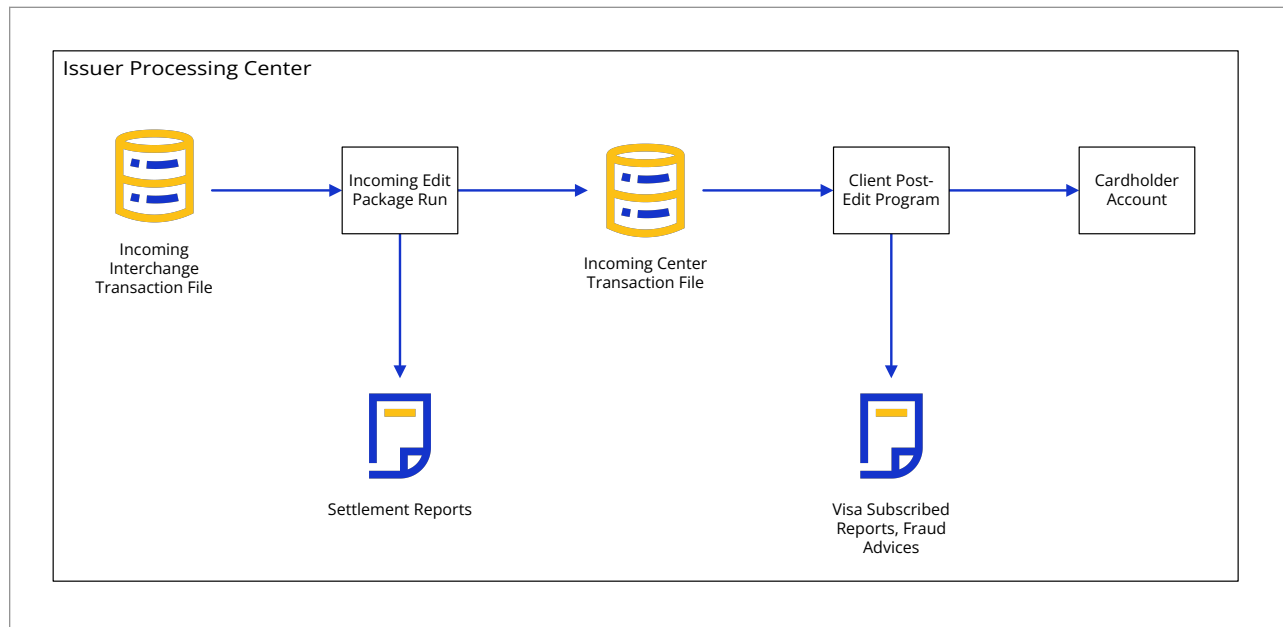
Related Information:

[Clearing Settlement Window Closures](#)

BASE II Issuer Transaction Cycle

The issuer's incoming Interchange Transaction File (ITF) is fed through the Incoming Edit Package Run from which clients can extract their own settlement reports. The incoming ITF, after editing, becomes the incoming Center Transaction File (CTF), which is fed into the issuer's post-edit program, for the posting of financial transactions to the cardholder account, as well as other non-financial information, such as fraud advices or Visa subscribed reports.

Issuer Processing Flow



The issuer or its processor reviews the clearing batches, removes the hold from the cardholder's funds, and posts the amount of the transaction to the cardholder's account.

At the end of each monthly billing cycle, the issuer prepares a statement for the cardholder, listing all activity on the account during the billing period. When the cardholder pays his or her bill to the issuer, the transaction cycle is complete.

Chapter 3

BASE II Clearing System Interchange Processing

BASE II Clearing System uses interchange processing to provide the electronic movement of transaction data between Visa's acquirers and issuers.

Outgoing interchange – Transaction data moving from client processing centers to the VIC.

Incoming interchange – Transactions and reports moving from the VIC to client processing centers.

When the terms “incoming” and “outgoing” are used, it is always from the client's viewpoint: outgoing means the client is sending a file to Visa and incoming means the client is receiving a file from Visa.

BASE II Financial vs. Non-Financial Interchange

BASE II Clearing System handles the exchange and transport of information for many Visa services, including financial and non-financial interchange.

About 60% of the transactions flowing through VisaNet are financial while the other 40% are non-financial. BASE II handles the exchange and transport of information for many Visa services, supporting these types of financial and non-financial interchange:

- **Originals** – Include purchases, credit vouchers issued after the return of merchandise, and cash disbursements, original credits, account funding, and some forms of money transfer.
- **Dispute Financials** – Transactions returned by the issuer to the acquirer because of disputes or infractions of processing rules.
- **Dispute Response Financials** – Transactions that are resubmitted to the issuer's processing center.
- **Reversals** are transactions that negate originals, dispute response financials, or dispute financials, in order to correct transactions that were duplicated in error.
- **Returned items** – Transactions sent back to the initiating processing center because the VIC edit function detected errors.
- **Reclassification advices** – Notices to sending clients regarding transactions that were processed (not returned) but did not perform as anticipated, and how they deviated from expectation.
- **Fee collections and funds disbursements** – Transfers of monetary credit from Visa to a client, or from one client to another for agreed-upon services (for example, card recovery rewards).
- **Non-financial data** – Includes fraud advices, batch acknowledgments, settlement data and reconciliation information, requests and confirmations of originals or copies, free-format messages, V.I.P. authorization advice records, informational reports, reclassification advices, and other data.

BASE II Transaction Codes

BASE II Clearing System uses transaction code (TC) numbers to indicate the type of transaction being sent or received.

Each type of transaction is identified by a transaction code (TC) number that indicates the type of transaction being sent or received.

This table lists a definition, a business case example, and, if it is a financial transaction, the financial flow between issuers and acquirers, for each transaction code (TC).

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 00 – Short Block	A TC 00 is used for filling a block in a file. A block is 12 records. A block can contain multiple transactions, but a single transaction can't span two blocks. Example: Transactions 1, 2, and 3, each contains multiple transaction records and go into one block but there is not enough space in the block for the four TCRs in the next record. The fourth transaction goes into the next block, and the first block is completed with one or more short blocks. Financial Flow: None
TC 01 – Returned Credit	A TC 01 is an incoming transaction that is generated by the BASE II Clearing System when a client sends in a transaction that would have resulted in a credit to them, but it failed the system edits because there is something wrong with the transaction. Visa returns the original transaction as a TC 01. Within a TC 01, TCRs 0 through 8 contain the contents of the returned item, and TCR 9 contains information identifying the original transaction (that is, the original destination, original source, original transaction code, why it got returned, what batch it came in on, etc.). Example: An acquirer sends purchase transaction (TC 05) that is returned as a TC 01 Returned Credit because it resulted in a credit to the source of the transaction. Financial Flow: Transaction – Visa to client Settlement – Debit to client Fee – None
TC 02 – Returned Debit	A TC 02 is an incoming transaction that is generated by the clearing system when a client sends in a transaction that would have resulted in a debit to them, but it failed the system edits because there is something wrong with the transaction. Example: An acquirer sends a credit voucher that does not pass an edit. Visa returns the transaction as a TC 02 Returned Debit. Financial Flow: Transaction – Visa to client Settlement – Credit to client Fee – None
TC 03 – Returned	Non-financial transaction. A TC 03 is an incoming non-financial transaction that failed an edit and is returned. Non-financial transactions include data capture advices, fraud advices, and ICS input requests. Example: A card issuer in the US receives an application for a credit card and sends in to Visa a TC 30 Issuer's Clearinghouse Service (ICS) query to determine the credit worthiness of the applicant. Visa is unable to forward the query on to ICS when the TC 30 fails an edit of a mandatory field and so Visa returns the transaction as a TC 03 to the issuer. Financial Flow: None

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 04 – Reclassification Advice Transaction	A TC 04 is an incoming transaction that notifies the client that a transaction was reclassified and did not receive the fee the client thought it would get. When a client submits a transaction for clearing, they indicate the interchange reimbursement fee for which they believe the transaction will qualify. If it does not qualify for the anticipated fee, Visa generates a TC 04. Within a TC 04, TCRs 0 through 7 contain the contents of the submitted item, and TCR 9 contains information identifying what fee they asked for, what fee they got, and the reason why they did not receive the anticipated fee. Reclassification advices are a configuration option: clients select whether they want to receive them, not receive them, or have the transaction sent back for correction if Visa reclassifies it. Example: A client submits a transaction that they believe qualifies for the Passenger Transport fee program but does not include the required information detailing each leg of the journey. Consequently, they do not get the fee and Visa sends them a reclassification advice informing them why they did not qualify. Financial Flow: None.
TC 05 – Sales Draft or Dispute Response Financial	Submitted by an acquirer, a TC 05 debits the issuer for the purchase of goods or services. If the TC 05's Usage Code is 1, this is an original transaction (that is, the first time the acquirer has submitted the transaction). If the issuer sends the transaction back, but the acquirer determines it is valid and resubmits it, the acquirer changes the Usage Code to 2 and the TC 05 becomes a Sales Draft Dispute Response Financial. If the issuer still disputes it, then the transaction goes into arbitration. Example: An acquirer submits a TC 05 Sales Draft for \$41.95, with a Usage Code of 1. BASE II processes it as an original transaction and forwards it to the issuer, who accepts it. Financial Flow: Transaction – Acquirer to issuer Settlement – Issuer to acquirer Fee – Acquirer to issuer

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 06 – Credit Voucher (aka Merchandise Return)	Submitted by an acquirer, the TC 06 is a cardholder or merchant initiated credit voucher, typically for returned merchandise. However, a TC 06 can also be an original credit, where the cardholder's account is receiving a credit without merchandise being involved, such as online gambling winnings or a gift from someone (e.g., Aunt Millie has her bank credit your Visa account for \$100 as a birthday gift to you). The TC 06 is an original transaction: on merchandise returns, Visa does not link the credit voucher with the sales draft. Credit vouchers can be re-presented. Example: A cardholder orders an orange sweater from an online merchant and when the merchant sends the order, the cardholder hates the color of the sweater. The cardholder sends the ugly orange sweater back, asking for a refund, and the merchant issues a credit voucher. Financial Flow: Transaction – Acquirer to issuer Settlement – Acquirer to issuer Fee – Issuer to acquirer
TC 07 – Cash Disbursement	TC 07 can be either a cash disbursement from an ATM or a manual cash disbursement from a bank, cruise ship, hotel, or other vendor. Example: An American cardholder vacationing in Cancun, Mexico, runs short on cash and uses her Visa card to get \$200 in pesos from the hotel's front desk. The transaction is submitted as a Cash Disbursement. Financial Flow: Transaction – Acquirer to issuer Settlement – Issuer to acquirer Fee – Issuer to acquirer
TC 10 – Fee Collection	TC 10s are bank-to-bank transactions or Visa-to-bank transactions that result in a credit to the originator and a debit to the destination. It is used by Visa to settle fees such as monthly transaction charges, quarterly fees, and membership fees. Financial Flow: Transaction – Source to destination Settlement – Destination to source Fee – None

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 15 – Sales Draft Dispute Financial	TC 15s are the result of a sales draft that went from the acquirer to the issuer, but the issuer is now requesting a dispute financial because something wrong with the sales draft. There are many things that can trigger a dispute financial, such as the transaction was above the merchant's floor limit which the merchant did not authorize, or a cardholder disputing the transaction, saying they never purchased the item in question (See <i>Visa Core Rules and Visa Product and Service Rules</i>, Dispute Resolution Rules, for full list of dispute reason codes and their specific conditions.) Dispute financials are originated by a card issuer and are destined to an acquirer. Either the entire amount of the transaction can be disputed or a portion of it. In a multicurrency situation, the dispute financial is valued using the currency exchange rates the day the dispute financial is processed, not the date the sales draft was processed. Despite a fluctuation the exchange rates, the cardholder is always made whole with the amount that was in the original sales draft. Example: The cardholder purchased sheets and towels from an Internet merchant. The cardholder's account is charged but the merchandise is never received. The issuer is submitting a dispute financial to the acquirer for the amount of the towels and sheets. Financial Flow: Transaction – Issuer to acquirer Settlement – Acquirer to issuer Fee – Issuer to acquirer
TC 16 – Credit Voucher Dispute Financial	TC 16s are the result of a credit transaction that went from the acquirer to the issuer, but the issuer is now requesting a dispute financial because it was not able to post to the cardholder's account. The issuer sends a dispute financial to the acquirer using a TC 16. This situation does not occur very often. Example: The issuer received a credit transaction and was not able to post it to the account because the account had been closed. Financial Flow: Transaction – Issuer to acquirer Settlement – Issuer to acquirer Fee – Acquirer to issuer
TC 17 – Cash Disbursement Dispute Financial	TC 17s are the result of a cash disbursement (either ATM or manual) that went from the acquirer to the issuer, but the issuer is now requesting a dispute financial. The issuer sends a dispute financial to the acquirer using a TC 17. Either the entire amount of the transaction can be disputed or a portion of it. Example: The cardholder requested \$100 from an ATM but only received \$80. The cardholder's bank requests a \$20 Cash Disbursement Dispute Financial. Financial Flow: Transaction – Issuer to acquirer Settlement – Acquirer to issuer Fee – Acquirer to issuer

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 20 – Funds Disbursement	TC 20s are bank-to-bank transactions or Visa-to-bank transactions that result in a debit to the originator and a credit to the destination (for example, to distribute rewards for card recovery). The opposite of TC 20s are TC 10s Fee Collection. Example: A card that has been reported stolen by Bank A is confiscated by Bank B's merchant when someone tries to use it in his or her store. Bank A uses a TC 20 to send the \$50 reward to Bank B, which forwards it to the merchant. Financial Flow: Transaction – Source to destination Settlement – Source to destination Fee – None
TC 25 – Sales Draft Reversal	TC 25s are submitted by an acquirer to reverse the result of a processing error, such as the acquirer accidentally sending through the original sales draft twice. The acquirer sends in a TC 25 Sales Draft Reversal, totally undoing the TC 05 debit to the cardholder. In a multicurrency situation, the reversal is valued using the currency exchange rates from the day the sales draft was processed. Example: A new sales clerk mistakenly charges a cardholder twice for the same item. The merchant's bank sends in a TC 25 to reverse the second sales draft. Financial Flow: Transaction – Acquirer to issuer Settlement – Acquirer to issuer Fee – Issuer to acquirer
TC 26 – Credit Voucher Reversal	TC 26s are submitted by an acquirer to reverse the result of a credit voucher processing error, such as a cardholder being credited twice for one returned item. It totally undoes the effects of a doubly-processed TC 06 Credit Voucher, resulting in a debit to the cardholder. In a multicurrency situation, the reversal is valued using the currency exchange rates from the day the credit voucher was originally processed. Example: An online gambling casino credits a cardholder's account with \$100 in winnings when the cardholder actually won \$10. The acquirer's bank submits a TC 26 Credit Voucher Reversal to reverse the \$100 credit and TC 06 Credit Voucher for the \$10. Financial Flow: Transaction – Acquirer to issuer Settlement – Acquirer to issuer Fee – Acquirer to issuer

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 27 – Cash Disbursement Reversal	TC 27s are submitted by an acquirer to reverse the result of a cash disbursement error, such as an ATM transaction that was inadvertently sent to Visa twice. Example: An American cardholder vacationing in Cancun, Mexico uses his Visa card to obtain \$200 in pesos from the front desk of their hotel, but the hotel mistakenly puts the transaction through twice. The hotel manager discovers the error and asks the hotel's processor to send in a TC 25 to reverse the second cash disbursement. Financial Flow: Transaction – Acquirer to issuer Settlement – Acquirer to issuer Fee – Acquirer to issuer
TC 30 – ICS Input	TC 30s are used by issuers to submit card applicant information to Visa, which forwards it to a third party to verify the applicant's credit card history and credit information. In the U.S. the third party doing the credit check is the Issuer's Clearinghouse Service (ICS), and outside the U.S., it is the National Application Review Service (NARS). The third party responds back to the issuer by sending a TC 31 ICS Response Transaction to Visa, which forwards it to the issuer. Example: An issuing bank in the US sends Visa a TC 30 containing information given to them by an applicant for a Visa card. Visa forwards it to ICS for verification. Financial Flow: None
TC 31 – ICS Response Transaction File	Used by either ICS or NARS to send credit history information about a Visa card applicant to an issuing bank. It is sent in response to a TC 30. (See TC 30 – ICS Input information above.) Financial Flow: None
TC 33 – Multi-Purpose Message Transactions	TC 33 is used by Visa to send a variety of non-financial messages to clients, including updates to the Plus Identifier tables, combined Visa/Plus routing table, Visa Transaction Routing tables, Interlink Routing tables, Visa Electron Routing Table, Mastercard Routing, Cirrus Routing, Armed Forces Financial Network, ATM No Surcharge Routing File, VisaVue Solution Series data, POS Acceptance Types Files, enrollment information for Visa Extras, and Visa Extra Daily Point Balances. Example: Visa sends a TC 33 to an issuer to notify them that one of their cardholders has enrolled in the Visa Extras point program via the Visa Extra's website. Financial Flow: None

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 35 – Sales Draft Dispute Financial Reversal	TC 35 reverses a TC 15 Sales Draft Dispute Financial. On the TC 15 the issuer requested a dispute financial but by submitting the TC 35 is now asking for that to be reversed, typically because of a duplicate file. Example: An issuer discovers that two dispute financials got sent in for the same sales draft and corrects the mistake by sending a TC 35 to the acquirer to reverse one of the dispute financials. Financial Flow: Transaction – Issuer to acquirer Settlement – Issuer to acquirer Fee – Issuer to acquirer
TC 36 – Credit Voucher Dispute Financial Reversal	TC 36 reverses a TC 16 Credit Voucher Dispute Financial. On the TC 16 the issuer requested a dispute financial but by submitting the TC 36 is now asking for that to be reversed, typically because of a duplicate file. This situation occurs rarely. Example: An issuer discovers that two dispute financials got sent in for the same credit voucher and corrects the mistake by sending a TC 36 to the acquirer to reverse one of the dispute financials. Financial Flow: Transaction – Issuer to acquirer Settlement – Acquirer to issuer Fee – Issuer to acquirer
TC 37 – Cash Disbursement Dispute Financial Reversal	TC 37 reverses a TC 17 Cash Disbursement Dispute Financial. On the TC 17, the issuer requested a dispute financial but by submitting the TC 37, is now asking for that to be reversed, typically because of a duplicate file. Example: An issuer discovers that two dispute financials got sent in for the same cash disbursement and corrects the mistake by sending a TC 37 to the acquirer to reverse one of the dispute financials. Financial Flow: Transaction – Issuer to acquirer Settlement – Issuer to acquirer Fee – Issuer to acquirer
TC 38 – Copy Request Service/Chargeback Documentation Automation Service Message	Visa Internal Use - TC 38 VDAS (for legacy chargebacks & representments), TC 38 VCRFS transactions (RFC) advices), TC 38 Non-Fulfillments are exchanged between RSI and VROL.
TC 39 – Automated Copy Fulfillment	Visa Internal Use - TC 39 Fulfillments are exchanged between RSI and VROL.

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 40 – Fraud Advice	The TC 40 serves as an envelope for an issuer or an acquirer to report fraudulent cardholder or merchant transactions to the System and Visa's Fraud Reporting System. It is also used for incoming optional print image reports to clients. Clients must electronically report all fraud to Visa, but fraud reporting requirements may be affected by specific programs, regional initiatives, or both. When Visa's Fraud unit receives a TC 40, it edits the information to determine if it is complete. If the client has opted for daily reporting, the Fraud unit sends daily a Visa-generated TC 40 Transaction Response Record contain a report image that lists for the sender which of their fraud advices were accepted, which were accepted but need additional information and resubmission, and which were rejected and require correction and resubmission. If the client has selected weekly reporting, they receive weekly a TC 45 print image report. Example: An issuing bank gathers the required fraud advice data regarding a purchase made with a stolen credit card and sends a TC 40 to Visa to add the fraudulent transaction to the Visa Fraud Master File. Financial Flow: None
TC 42 – Merchant File Updates	The TC 42 serves as an envelope for acquirers to report to Visa required merchant update information. Visa maintains a large database of information on merchants who accept Visa cards. This information is used for tracking merchant compliance to the <i>Visa Core Rules and Visa Product and Service Rules</i>, and for maintaining the National Merchant Mailing File used for distributing the Card Recovery Bulletin. Example: An acquiring bank gathers the required information on his new merchants and sends a TC 42 to Visa to add the data to the National Merchant Mailing File. Financial Flow: None Note: TC 42 and TC 43 transactions are no longer valid BASE II transaction types. Transactions will be returned with return reason code of G7.
TC 44 – Collection Batch Acknowledgment	TC 44s are Visa-to-client transactions containing information about the batches that the client has sent to Visa (i.e., batch number, number of transactions in the batch, dollar amount of the batch, disposition of the batch, etc.). This information is used by the incoming Edit Package to match what was sent by the client against what Visa acknowledges receiving, and then produce reports for the client. Example: A client receives a TC 44 from Visa reporting a batch was received and verifying the information in the batch. Financial Flow: None
TC 45 – General Delivery Report	TC 45s are used by Visa to send to clients a variety of printable reports to clients (settlement, fraud, etc.). Example: A small credit union receives a TC 45 of settlement information from Visa. They print out the report and a clerk reconciles the figures against the bank's records. Financial Flow: None

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 46 – Member Settlement Data	TC 46s are used by Visa to send to members settlement data in machine-readable format, if the member wants to reconcile with Visa in an automated way. Example: A large bank receives a TC 46 of settlement information from Visa. The information is fed into a program, which reconciles the figures against the bank's records. Financial Flow: None
TC 47 – Report Generation.	TC 47s are used by Visa to transmit to acquirers a report on authorization processing activity. Financial Flow: None
TC 48 – BASE I Advice	TC 48s are used to inform clients of online authorizations performed on their behalf by Visa. When a Visa card issuer is unavailable to respond to an authorization request within a defined period (typically 10 to 20 seconds), Visa can perform Stand-In Processing (STIP) for them, responding to the request based on criteria set by the issuer. The issuer can also choose to use stand-in processing for all transactions under a certain dollar limit. If STIP processing was performed, the system sends a TC 48 BASE I advice to the issuer so that they can update the cardholder's "open to buy" balance. In rare instances, TC 48s are sent by acquirers to notify issuers when a chip card interacting with a terminal in an offline environment decides to decline a transaction. Issuers have the option of receiving TC 48s through BASE II or having them held until they retrieve them via the online system. Example: A client's system goes down for a minute and during that time Visa performs Stand-In Processing on two transactions. Visa sends two TC 48s to the client to inform them of the authorizations performed on their behalf. Financial Flow: None
TC 50 – Free Text Message	TC 50s are used by Visa and clients to send a wide variety of non-financial, unformatted, text messages. TC 50s are used heavily with commercial credit cards to include additional data that is too large to be included in the financial message. TC 50s can be chained together to create large messages (e.g., on a purchasing card, TC 50s can be strung together to add extensive invoice line item information). Example: In an airline transaction, the data describing up to 32 legs of a particular ticket can be included in a TC 50. Financial Flow: None
TC 52 – Request for Photocopy or Substitute Draft	TC 52s are requests for a copy of the original sales draft. An image of the original transaction is required in the case of a dispute. A TC 39 is generated in response to the TC 52 and provides an electronic image of the original sales draft. Example: An issuer whose cardholder is disputing a charge sends in a TC 52, requesting a copy of the original sales draft. The acquirer responds with a TC 38 and a TC 39, showing an electronic image of the signed sales draft. Financial Flow: None Note: TC 52 transactions (Retrieval Requests) are no longer valid BASE II transaction types. October 2021 TC 52 transactions will be returned with reason code 02.

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 54 – Table Update Data	In addition to program updates, Edit Package generates and distributes updates to BASE II VID and ARDEF tables and to other value tables and business rules via TC 54. The Edit Package software code updates may be delivered via TC 54s or made available through Visa Access. If updates were delivered via TC 54s, the system automatically applies the updates to the Edit Package software during an incoming run. Example: A client whose ARDEF files have become corrupted requests a replacement copy. Visa sends the client the replacement files via a string of TC 54s. Financial Flow: None
TC 55 – RCRF (Regional Card Recovery Files Update).	TC 55s are an electronic version of the Card Recovery Bulletin, informing clients of lost and stolen cards and blocked BINs. They provide acquirers with means of identifying cards that issuers no longer honor and would like confiscated. If a cardholder reports a card as lost or stolen, their issuer notifies Visa of the account number via their online connection. They also give the card a pick-up code, and indicate in which regions they want the card number blocked for transactions. Visa maintains a cardholder database and extracts information to create the Card Recovery Bulletin in paper, CD-ROM, and TC 55s. In addition to listing which card numbers have been lost or stolen, TC 55s also report BIN numbers that have been blocked (any card issued by that BIN is no longer valid). Example: An issuer sends file updates containing stolen and lost card numbers to Visa. The Card Recovery Service uses this data to create the Card Recovery Bulletin, which they distribute worldwide via paper copies, CDs, and TC 55s. Financial Flow: None
TC 56 – Currency Conversion Rate Update	TC 56s are used by Visa to transmit daily updates of the currency conversion rate file to the processing center's of subscribing clients. Subscribing centers are provided with the same Currency Conversion Rate information that VisaNet Interchange Centers use. The contents of the TC 56 file are confidential and may be used only by Visa and its clients. Example: A client in Vancouver, Canada, a multi-currency environment, receives daily TC 56s to keep informed of the Visa buy and sell rates of the currencies they deal in. Financial Flow: None

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 57 – Data Capture Transaction Advice	TC 57s are used by Visa to send acquirers information about a particular transaction so they can submit the transaction as a sales draft. This typically is needed when an acquiring bank signs up merchants but outsources the Point of Sale terminal network to a third-party Data Capture Service. The third-party contractor collects the data and transmits it to Visa, which Visa forwards on to the acquirer using TC 57s. Each TC 57 contains information about a specific transaction. The information is formatted by the acquirer into a TC 05 Sales Draft, and submitted to Visa. TC 57s are also used to transmit to acquirers non-Visa card transactions captured by Visa terminals. Example: A third-party Data Capture Service sends to Visa transactions, including Mastercard and American Express transactions, captured on Visa terminals at specific merchant. Visa captures the format the merchant batch data into TC 57s and forwards them to the appropriate acquirer. Financial Flow: None
TC 58 – National Settlement Advice	TC 58s are used to send clients in National Settlement services their settlement information. In addition to the International Settlement Service, Visa also allows National Settlement, which enables the clients within a country to settle in their own settlement service all domestic transactions (that is, transactions where the issuer, acquirer and merchant are all in the same country are settled within the country without being converted one of Visa's 28 settlement currencies). Example: Visa generates TC 58s for the participants in the settlement service and forwards it to the appropriate clients of that service. Financial Flow: None
TC 59 – Interface Settlement Advice	TC 59 records are used to advise non-Visa card issuers of a transaction processed by a Visa acquirer or data capture provider. The TC 59 records are created from merchant batch data and are submitted to Visa for delivery to the non-Visa card issuers. The non-Visa card issuers use the TC 59 records for their own clearing and settlement. Example: A third-party Data Capture Service sends to Visa the American Express transactions they have captured on Visa terminals. The acquirer formats the merchant batch data into TC 59s and forwards them to American Express. Financial Flow: None
TC 90 – File Header	TC 90s contain non-business information used to identify when a file was created and who created it. The File Header identity information includes a security code as well as a field indicating whether it is a test file or production file. For outgoing files, the client loads files with TC 90s attached onto their VAP, EA server, or other connectivity device and Visa collects the files. When the files are fed into BASE II for processing, the TC 90 identifies the client who sent the file. TC 90s are also used by Visa to forward files to clients. On incoming files, the TC 90s also include the settlement date. Financial Flow: None

Transaction Codes, Definitions, and Examples

Transaction Code and Type	Definition, Example, and Financial Flow
TC 91 – Batch Trailer	TC 91s summarize what is in a batch: how many records, how many transactions, a hash total for all the amounts. This control information is used by Visa to validate batches. When Visa runs a batch, it compares its totals with the totals in the batch trailer and if the figures don't match, it rejects the batch. Financial Flow: None
TC 92 – File Trailer	TC 92s summarize what is in an entire file: how many records, how many transactions, a hash total for all the amounts. This control information is used by Visa to validate the file. Financial Flow: None

BASE II Transaction Component Records

A transaction is typically composed of multiple Transaction Component Records (TCRs), each of which contains 168 bytes of data. It is recommended that both Outgoing and Incoming Interchange Transaction Files (ITFs) contain up to 999 TCRs in a batch.

The first TCR in a transaction, TCR 0, tells basic information about the transaction, such as what the source and destination were. TCR 0 is a mandatory part of every transaction, but all the subsequent TCRs are optional.

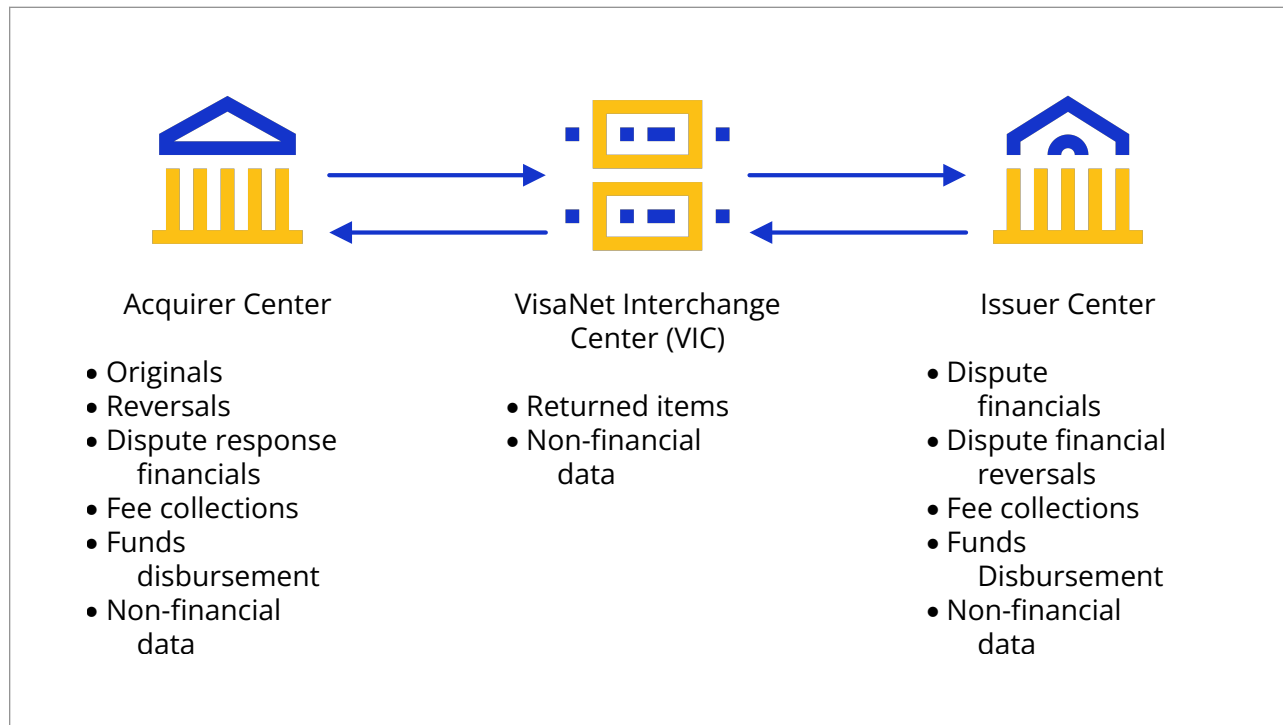
Transactions contain an average of two-and-a-half TCRs per transaction. Most TCRs have defined purposes (for example, on a TC 01 Returned Credit transaction, the TCR 9 within it carries information about the original transaction and the reason it was returned).

BASE II Outgoing Transaction Processes

BASE II Clearing System transactions can be initiated by an acquirer, issuer, processor, or Visa. Regardless of the originator, all outgoing BASE II transactions are processed using the same procedures: transactions are batched into a file and then sent to the BASE II System at the VIC.

This graphic illustrates the flow of data into BASE II.

Basic Data Flow Into BASE II



BASE II Outgoing Interchange

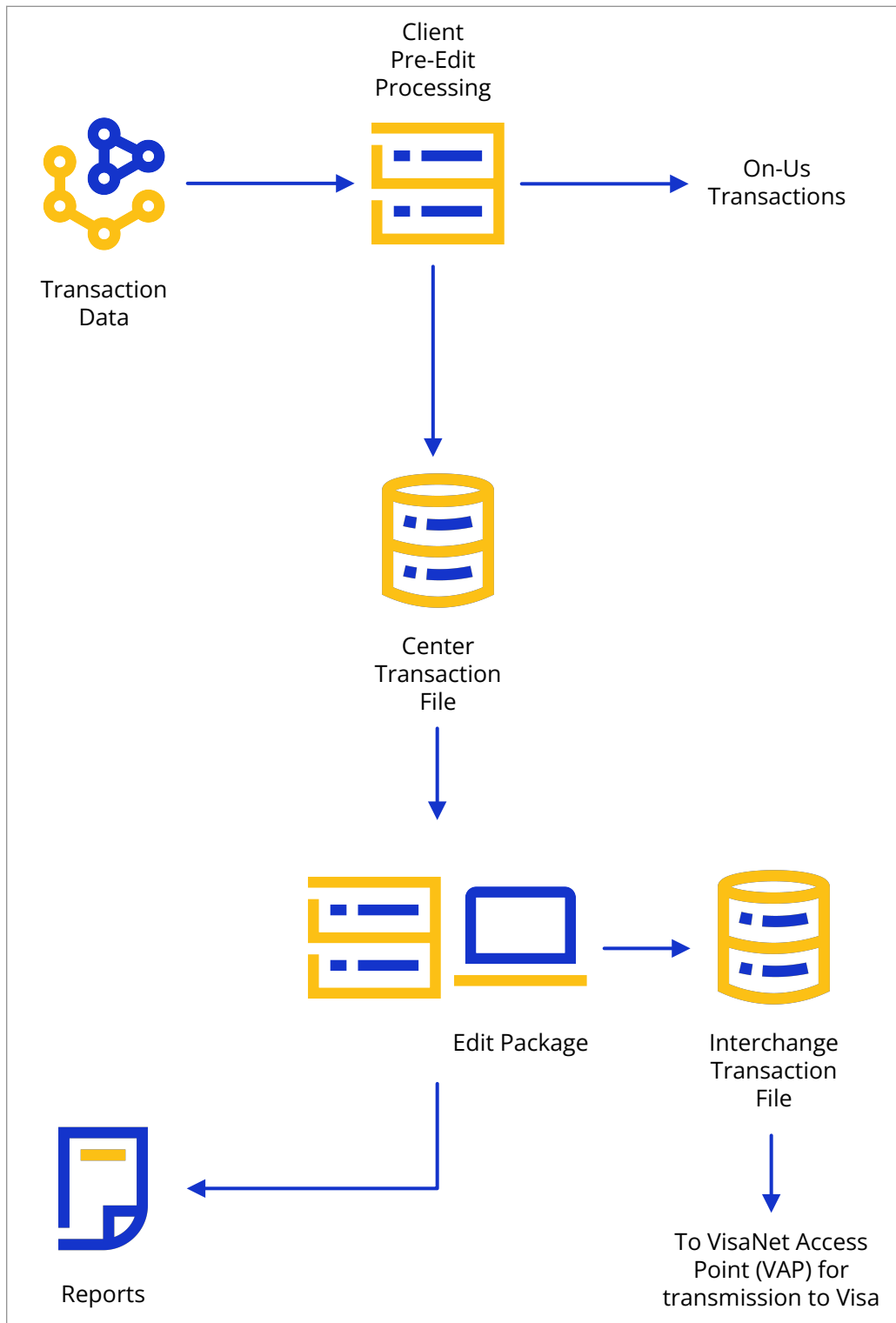
BASE II Clearing System uses the pre-edit software, the Center Transaction File (CTF), and the Edit Package to provide the outgoing interchange.

Transactions are first processed by each center's pre-edit software. Interchange transactions are placed in a Center Transaction File (CTF). This file is then processed through the Edit Package, which validates the information and ensures that it is free of obvious error.

Edit Package outputs Interchange Transaction Files (ITFs), which are transferred to a VAP, EAS, or other connectivity device, where they are held until a predetermined collection time. If they send through the US Message Gateway, the endpoint has a unique communications ID and a telephone number for its communication device that Visa dials out to. The transmission of the files is jointly managed by the device and the VIC, with little or no intervention by center personnel, required.

This graphic illustrates the flow of data through the outgoing interchange.

BASE II Outgoing Interchange



These sections describe the flow of data through the outgoing interchange.

Endpoint Authentication

When a file comes in, the VIC reads where the header record came from and performs a high-level edit to determine if this is a valid endpoint for this file: Does Visa recognize the endpoint number in the configuration? Does this endpoint process for this BASE II CIB? In other words, Visa authenticates the logical source of the communication.

Data Protection

Once interchange data is accepted by the VIC, it is protected from system failure. In the event of a failure, however, the BASE II System can restart processing where the failure occurred. If requested, client files can also be rebuilt and delivered electronically back to the processing center.

On-Us Transactions

“On-us transactions” are those in which the cardholder and merchant are serviced by the same financial institution. These transactions are not usually sent through BASE II, since they do not require clearing or settling with another client, but they can be sent if the client has a reason for doing so (for example, inclusion in a Visa sweepstake promotion). With on-us, clients make internal financial transfers between merchants and cardholders and apply the transactions immediately to their accounts.

Intra-Processor Transactions

In the U.S., *intra-processor transactions* are required to be sent to BASE II for risk management purposes. Intra-processor transactions involve two different clients that are serviced by the same processing center. As an option, intra-processor transactions may also be settled instead of being reported without settlement.

BASE II Incoming Interchange

BASE II Clearing System uses connectivity systems, an Interchange Transaction File (ITF), and the Edit Package to provide the incoming interchange.

The incoming interchange process begins when the VIC transmits interchange to the center’s VAP, EAS, or other connectivity device. Data is sent in the form of an Interchange Transaction File (ITF), which is transferred to the host and then processed through the Edit Package.

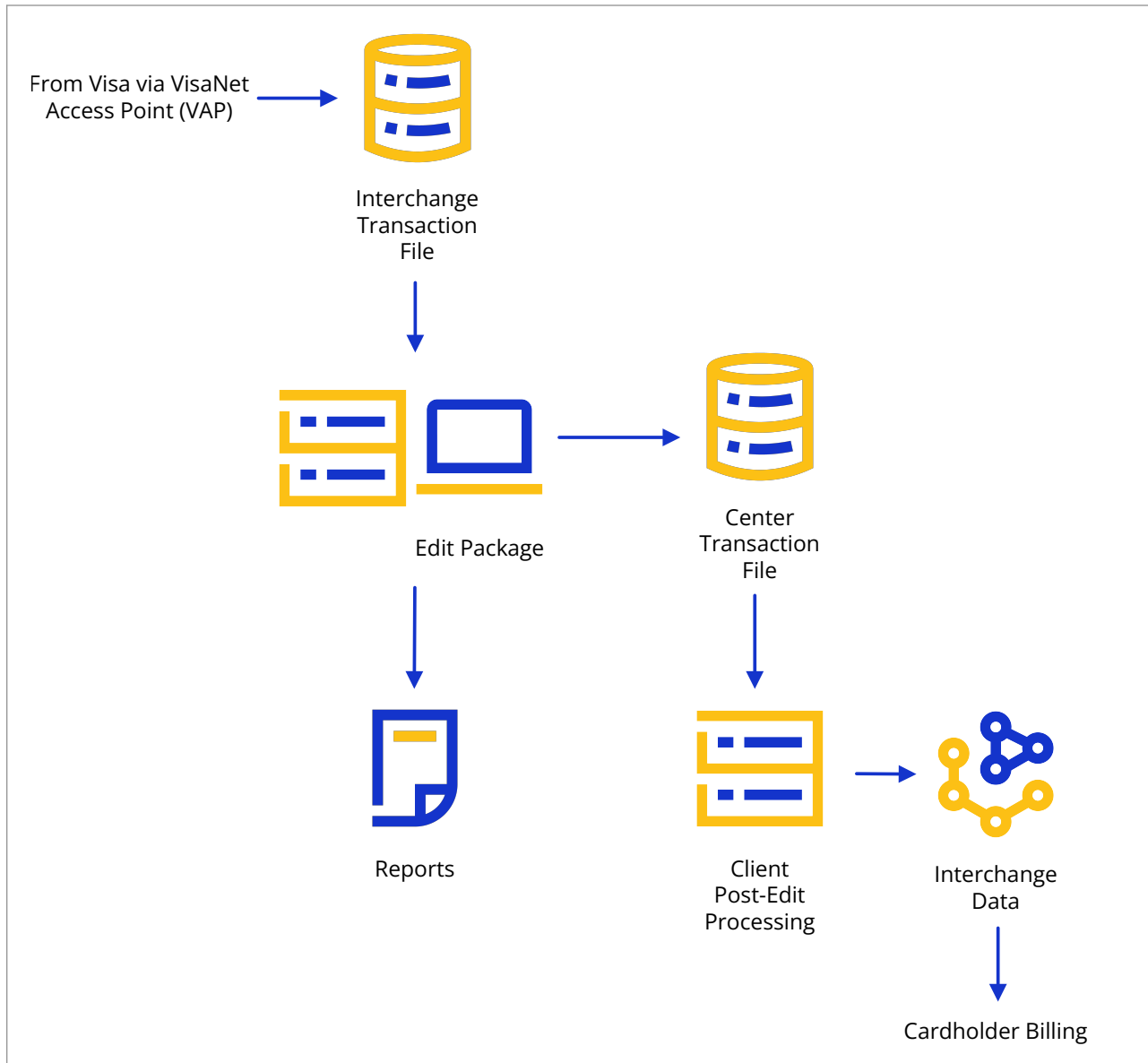
Incoming Edit Package files typically contain:

- Financial and non-financial interchange transactions.
- Settlement totals.
- Batch acknowledgments.
- BASE II settlement reports.
- Edit Package table update transactions.
- Routing tables.

During the incoming process, the Edit Package produces a Center Transaction File (CTF) containing all the incoming transactions destined for the clients serviced by the processing center. Reports for control and reconciliation are also produced.

This graphic illustrates the flow of data through the incoming interchange.

BASE II Incoming Interchange



BASE II Processing Schedule

BASE II Clearing System operates seven days a week. Each daily cycle goes through a collection, delivery, and settlement phase.

- **Collection** – Process of collecting BASE II interchange from across the globe (runs simultaneously with delivery). Collection times must be established for BASE II to connect with the client's VAP or other connectivity device for interchange. Some clients do not require collection times (i.e., Internal Visa Processors). To prevent files from being collected or delivered on a particular day of the week, the client selects that day of the week for “No Scheduling Activity.”
- **Delivery** – Process of sending interchange to its final destination, accompanied by reconciliation report data for the cleared and settled transactions. All processing centers receive an incoming interchange file seven days per week. BASE II automatically delivers post settlement. Delivery times allow for the scheduling of extra deliveries (see below) in addition to the end of the cycle file.
- **Settlement** – Process of determining the value of interchange, including any fees and charges. The result of settlement is the client's net position for a given cycle. During this process, settlement reports are generated.

BASE II Collection and Delivery Options

In addition to standard collection and delivery, the processing center offers a number of collection and delivery options.

Multiple Collection – This option allows clients to send outgoing interchange more than once during the cycle. By allowing transaction data to be split into multiple files that are collected at different times, processor workloads and schedules can be significantly eased.

Delivery Options – A client, or its third-party processor, can choose to have transaction data delivered before settlement is completed. Normally, transaction data and settlement reports are delivered simultaneously. However, a client may choose to have some data in advance, for example, in order to anticipate volume or to determine whether any transactions were rejected. Some of the delivery options are:

- **Scheduled Delivery** – The client receives its data through the BASE II System according to a predetermined schedule that best meets its needs and processing schedule (e.g., send what has been processed so far at 3 p.m, 5 p.m., etc.).
- **Delivery Based on File Size** – Data is delivered to the client based on volume processed (e.g., send files as soon as a specified number is completed).
- **Expedited Delivery** – This option enables immediate delivery of selected Customized Delivery files, regardless of the normal delivery schedule (e.g., send selected files immediately after collection and clearing of them has been completed).

- **Customized Information Delivery** – This option allows processors to customize their BASE II delivery files to a list of pre-defined file types. Delivering similar transactions in one individual file provides faster access to critical information (for example, separating financial transactions from non-financial transactions). BASE II processors may select as many Customized Delivery file types as needed from a list of dozens of types (e.g., Financial, Fraud, Back-Office, BASE I advices).
- **Split Routing** – This option allows BASE II processors to select alternate BASE II file delivery locations for certain transaction types. The Customized Delivery Service organizes delivery files by transaction type. With the Split Routing Service, processors can designate different BASE II delivery locations based on particular services that may be located at different processing centers.

Chapter 4

BASE II Clearing System Exception Processing

Exception Processing includes transactions that dispute, reverse, adjust, or correct "originals." They consist of all "non-originals" of purchase and cash disbursement transactions as well as administrative, file maintenance, fee-related, and reconciliation transactions.

An original refers to the original sales draft of a purchase or cash advance transaction. Non-originals refer to all follow-up transactions initiated after the original.

Exception item processing is a back-office function that supports credit and deposit access transactions. It provides a means for clients to correct transaction discrepancies. Exception items are follow-up messages between issuers and acquirers. Although exception messages are sent electronically, exception processing requires the manual support of client staff. In BASE II, there are two ways this is done: dispute financials and dispute response financials.

BASE II Dispute Financials

Dispute financials are initiated by an issuer to return a disputed transaction to the acquirer. There are a number of reasons why a transaction is returned. Some common ones include: card validity, authorization error, the cardholder disputes participating in the transaction, and processing errors.

There are a number of ways a dispute financial is initiated, depending on the reason for the dispute. Typically, it is the reverse of a transaction: the issuer sends a message to VisaNet submitting a dispute financial with a reason code that explains the type of dispute being submitted. VisaNet forwards this to the acquirer for resolution.

Another method of initiating a dispute transaction is through a service called Visa Resolve Online (VROL). VROL is a service available to issuers for initiating a dispute financial. The issuer connects to the service through an Internet connection, enters the transaction information, and the system goes to a database and retrieves the original transaction and populates the fields for the dispute financial. As the issuer enters the dispute reason codes online, the dispute financial is edited against the original transaction.

EXAMPLE

A cardholder orders merchandise over the internet. The amount of purchase appears on his credit card statement, but goods are never received.

The cardholder attempts to resolve this with the merchant and is unsuccessful because the merchant is unable or unwilling to provide the merchandise as promised.

The cardholder contacts the issuer, and provides documentation proving that an attempt to resolve the transaction with the merchant was unsuccessful.

The issuer creates a dispute financial transaction and sends it to Visa. Visa processes the transaction, assesses fees, and clears and settles it so the cardholder can receive a refund.

Visa forwards the dispute financial to the acquirer, who, after determining that the dispute financial is valid, collects the money back from the merchant.

See BASE II Clearing System Services and Visa Core Rules and Visa Product and Service Rules.

BASE II Chargeback Services

BASE II Clearing System uses chargeback services to reduce the time and money expended by issuers to research and process chargebacks.

The Chargeback Reduction Service (CRS) validates chargebacks, copy request and fulfillment transactions, and sales drafts. It then returns those that do not meet validation criteria and provides clients with helpful information about items that may be disputed.

BASE II Dispute Response Financials

BASE II Clearing System uses dispute response financials to resubmit a transaction when a dispute financial proves invalid.

A dispute response financial is a clearing record that an acquirer presents to an issuer through interchange after a dispute financial.

An acquirer, having already presented the original transaction to the issuer, resubmits the transaction to the issuer as a dispute response financial when a dispute financial proves invalid. An acquirer can submit one dispute response financial; if the dispute response financial is not valid, the dispute goes to arbitration.

Chapter 5

BASE II RSI Jurisdiction Rules

BASE II Clearing System uses the location of a transaction's issuer, acquirer, and merchant to determine a transaction's fees and associated edits and charges.

In order to determine a transaction's fees, edits, and charges, BASE II must first establish the transaction's jurisdiction. The location of a transaction's issuer, acquirer, and merchant drives how BASE II processes it. The system determines if the transaction is domestic, regional, or interregional:

- **Domestic Transaction** – Transaction that occurs within a single country or zone and receives fees defined by the clients within that country. A zone is a sub-category of region that is synonymous with country (for example, Canada and the U.S. region are each zones).
- **Regional Transaction** (or Intraregional Transaction) – Transaction that occurs between two countries in the same Visa region and receives fees defined by that region's Board.
- **Interregional** – The transaction occurs between two regions and receives fees agreed upon by Visa's International Board.

There can be up to four different jurisdictions per transaction:

- One for edits
- One for fees (costs that flow from client to client)

- One for charges (costs that flow from clients to Visa)
- One for disputes (rights are by merchant)

BASE II Jurisdiction Processing

BASE II Clearing System uses transaction-related information to determine the jurisdiction of a transaction.

There are many ways to determine the jurisdiction of a transaction.

- Issuer's region and country: uses card range to define where the issuer is located.
- Merchant's region and country: uses merchant's country code to determine where the merchant is located.
- Acquirer's region and country: uses the acquiring identifier that sent in the transaction to define where the acquirer is located.

RSI runs the above information through a series of tables and gets one line defining the jurisdiction.

RSI Jurisdiction Rules

This table contains the jurisdiction rules used for various types of domestic, regional, and interregional transactions..

RSI Jurisdiction Rules

Jurisdiction	Issuer	Acquirer	Merchant	Merchant Category Code
Domestic Air	In same country/zone as acquirer	Anywhere	In same country/zone as issuer	Airline
U.S. Domestic	In U.S. region/zone	In U.S. region/zone	In any region/zone	Not airline
Canadian Domestic	In Canada region/zone	In Canada region/zone	In Canada region/zone	Not airline
Asia Pacific Domestic	In same Asia Pacific region/zone as acquirer	In same Asia Pacific region/zone as issuer	In Asia Pacific region/country	Not airline
Latin America Domestic	In same Latin America region/zone as acquirer	In same Latin America region/zone as issuer	In Latin American region/country	Not airline

RSI Jurisdiction Rules

Jurisdiction	Issuer	Acquirer	Merchant	Merchant Category Code
Europe (EU) Domestic	In same Europe region/zone as acquirer	In same Europe region/zone as issuer	In same Europe region/zone as issuer	Not airline
CEMEA Domestic	In same CEMEA region/zone as acquirer	In same CEMEA region/zone as issuer	In same CEMEA region/zone as issuer	Not airline
Europe Cross-Border Acquiring	In EU region, and issuer zone is merchant zone	In EU region	In EU region, and merchant zone is same as issuer zone	Not airline
CEMEA Cross-Border Acquiring	In CEMEA region, and issuer zone is merchant zone	In CEMEA region	In CEMEA region, and merchant zone is same as issuer zone	Not airline
Regional (transaction not domestic)	In same region	In same region	In same region	Not airline
Regional Airline (transaction not domestic)	In same region as merchant but different country than merchant	In any region	In same region as issuer but different country than issuer	Airline
Interregional (transaction not regional or domestic)	In different region than acquirer	In different region than issuer	In any region/zone	Not airline

Chapter 6

BASE II Fees and Charges

The calculation of fees and charges is a major function of BASE II processing. In addition to moving transaction amounts, BASE II calculates and collects the compensatory payments clients make to each other and to Visa for risks borne by issuers and acquirers, and for a variety of administrative and processing services provided by Visa.

These payments fall into two categories:

- Interchange reimbursement fees between issuers and acquirers
- Transaction charges paid by clients to Visa

BASE II IRF Fees

Interchange Reimbursement Fees (IRFs) are transfer fees that are used to balance the benefits and costs of Visa card transactions between issuers and acquirers. Interchange fees compensate both parties for their role in providing a comprehensive payment service to cardholders and merchants.

IRFs are also used by Visa to encourage certain kinds of behavior in clients (for example, authorization of transactions and CPS participation) and to develop emerging products and technology (e.g., chip-capable terminals and contactless payments). Events such as movement in the economy or competitive rate changes may trigger interchange adjustments.

IRFs flow from one client to another based on the transaction.

- **Payment Transactions** – Fees flow from acquirer to issuer in credit transactions because the issuer is extending credit so the cardholder can make a purchase at the acquirer's merchant, and also to compensate the issuer for developing and maintaining the card payments mechanism that provides a service to merchants, acquirers, and cardholders.
- **Debit Transactions** – Fees flow from issuer to acquirer in ATM transactions because the acquirer has gone to the expense of buying the ATM and putting it on the network for the convenience of the issuer's cardholders.
- **Exception Processing** – On disputes, reversals, and other types of exception processing, interchange flows in reverse.

IRFs, which are levied on transactions flowing through VisaNet, provide a significant source of revenue to Visa's client banks. The fees are set by agreement between:

- The banks within a country for domestic transactions.
- The regions for regional transactions.
- By agreement between regions (represented by the Visa International Board) for inter-regional transactions.

BASE II supports over 11,000 legal, private and merchant agreements unrelated to jurisdiction.

BASE II IRF Calculation

BASE II Clearing System uses transaction characteristics to calculate the Interchange Reimbursement Fees (IRFs) for each transaction.

IRFs are calculated by BASE II during clearing and balanced during settlement. BASE II acts on a wide variety of transaction characteristics to accurately assign an appropriate interchange rate to each transaction. Some of the items which impact the fee are:

- **Brand** – What brand is the card
- **Product** – What is the product (check card, credit card, business card, prepaid, etc.)
- **Transaction Type** – What type of transaction (purchase: original, purchase: dispute response financial, credit: original, fee collection, etc.)
- **Network** – What network was the transaction sent on
- **Debit or Credit** – Is it debit or credit transaction
- **Settlement Service** – Which settlement service is used

- **Custom Payment Service (CPS)** – This program, which is used by the U.S. and Brazil, is structured to give acquirers economic incentives to provide data integrity between the authorization and clearing records and expedite the clearing of transactions.
 - Acquirers pay lower rates for transactions that meet CPS requirements.
 - Transactions that do not meet CPS qualifications are reclassified to a higher Electronic Issuers Reimbursement Fee (EIRF) and standard rates.
 - CPS requirements include consistency between the authorization and clearing messages (an inconsistent validation code means something has changed between authorization and clearing) and timeliness of clearing (if the transaction was not submitted within 3 days, there is increased risk that the account might have gone bad).
 - Specific merchant segments that qualify for CPS include supermarkets, hotels and car rentals, passenger transport, direct marketing, and automated fuel dispensers.
- **Merchant Verification Values (MVV)** – A unique identifier assigned by Visa to specific merchants to uniquely identify them on all transactions. The MVV is used to determine whether the transaction is eligible for a special fee program established by Visa.

See *Visa Core Rules and Visa Product and Service Rules*, *Visa U.S.A. Operating Regulations*, *Interlink Operating Regulations*, and *Plus Operating Regulations*.

BASE II Cashback Interchange Fees

Visa's cashback feature provides cardholders with added convenience by dispensing cash with a purchase transaction at the point of sale. When the cashback option is used in a purchase transaction (TC 05 – Sales Draft), the cashback field in the TC 05 indicates what portion of a transaction is cashback.

In Visa's US and Canada regions, the cashback field is informational only, telling the issuer that part of the transaction was cashback. Interchange is charged on the entire transaction. For example, a cardholder in Chicago purchases a \$1 candy bar and asks for \$20 cash back, with the total transaction coming to \$21. When the transaction amount (\$21) submitted by the merchant is cleared, BASE II determines the jurisdiction to be US domestic, ignores the cashback field, and charges the acquirer interchange on the entire \$21.

In certain countries within the Asia-Pacific, LAC, CEMEA, and VE regions, there are two variations on how interchange is handled on cashback:

- Interchange is charged only on the actual purchase amount, not on the cashback amount
- interchange is calculated on the cashback amount but flows in the opposite direction, thus giving an incentive to the merchant to provide the service.

For example, using variation one, if the cardholder purchases a \$1 candy bar in Hong Kong and asks for \$20 cash back, the total transaction still comes to \$21 but the interchange fee is only charged on \$1 of the total transaction. When BASE II processes the transaction, it determines

the jurisdiction is Asia Pacific, reads in the cashback field on the TC 05 that the cashback part of the transaction was \$20, and calculates interchange fees only on the actual purchase (\$1) rather than on the total transaction (\$21).

Argentina NNSS POS Cashback: for information on IRF processing and providing value added tax (VAT) on Argentina NNSS POS cashback transactions refer to your Visa representative.

BASE II Transaction Charges

Transaction charges are how Visa bills clients (or their third-party processors) for VisaNet processing services. Charges are specified in the regional Operating Regulations, and the region in which the acquiring/issuing identifier is located receives the charge.

All charges are billed monthly, except for International Service Assessment and International Outgoing Interchange charges, which are assessed daily. A tiered-charge structure is used (e.g., the first million transactions are charged one rate, the next million another rate).

Visa pricing varies depending on a number of factors, namely, the client's processing configuration, projected capacity, and actual transaction volume, as well as the product involved and any applicable incentive programs. The processing system (BASE II or Visa Global Billing Platform (VGBP)) assesses the charge to clients on behalf of the region and then releases the monies to the region. Visa uses TC 10 Fee Collection transactions, which result in a credit to the originator and a debit to the destination, to settle monthly transaction charges, quarterly fees, and membership fees.

BASE II Transaction Charge Types

BASE II Clearing System uses a number of factors to calculate the types of charges applied to each transaction.

Types of transaction charges:

- **International Service Assessment (ISA)** – If a transaction crosses a border (that is, the merchant and issuer are in different countries), Visa charges a percentage of the value of the transaction. The charge is paid by the issuer to issuer's region. The issuer region may choose to share with its issuers the monies collected or keep it. Some regions use the monies to fund special projects.
- **International Acquiring Fee (IAF)** – These charges are assessed by Visa to the acquirer of a cross-border (merchant and issuer are in different countries) transaction. IAF charges vary by Visa region. They are usually calculated as a percentage of the transaction amount and are collected through the daily settlement process or monthly client invoicing process. IAF fees apply to purchase and cash Original transactions and reversed on the Reversal transaction when the merchant is in the Asia-Pacific (AP), Latin America and Caribbean (LAC), Europe (EU), Central Europe/Middle East/Africa (CEMEA) and United States (US) regions.

- **International Service Assessment Fee (IASF)** – These charges are assessed by Canada region to the acquirer of a transaction when the transaction is between a merchant in Canada and an issuer outside of Canada. IASF is calculated as a percentage of the transaction amount and are collected through the daily settlement process. IASF fees apply to purchase and cash Original transactions and reversed on the Reversal transaction.
- **Acquirer International Service Assessment (AISA)** – These charges are assessed by Asia-Pacific (AP) and United States (US) regions to the acquirer of a cross-border transaction (merchant and issuer are in different countries) when the transaction is from a merchant in its region. They are calculated as a percentage of the transaction amount and are collected through the daily settlement process or monthly client invoicing process.
- **VAP Access Charges** – These charges are assessed to issuers, acquirers, and third-party processors directly connected to a VisaNet Access Point (VAP), the link to VisaNet. This monthly charge is comprised of a fixed access fee and a variable component based on actual transaction volume. It includes fees for settlement and reconciliation, transaction processing, and use of the V.I.P. System.

- **Processing Charges** – Visa assesses transaction switching charges from issuers for transactions processed through VisaNet. These charges, which may vary by transaction type (for example, returned items in the U.S. region are charged \$5 per transaction), are billed monthly. Generally, volume discounts apply to high-volume clients. Interlink processing charges (billed to Full Service clients only) follow a different structure. Visa also bills Interlink acquirers for transaction processing.
- **Administrative Charges** – Most administrative charges are billed monthly, though some are billed daily for U.S. endpoints. Administrative charges (applicable to both issuers and acquirers, except as noted) include:
 - **Application fees** – One-time charge to participate in a Visa payment program. Application fees are charged for Visa check card, Interlink, and Plus programs.
 - **Cardholder Database Residency Charges** – Charges applied to items retained on the Account Screen Authorization File (ASAF), the PIN verification file, and the address verification file. They are billed to issuers.
 - **Cardholder Database File Update Charges** – Charges assessed to issuers for updates to the Account Screen Authorization File (ASAF), the PIN verification file, and the address verification file.
 - **Merchant Central File Residency Charges** – These monthly charges apply to entries retained in the merchant central file and are paid by acquirers. There is no charge for adding a new account to the file.
 - **Merchant Central File Update Charges** – Assessed monthly to acquirers, these charges accrue for updates to the merchant central file.
 - **Certification Fees** – Applicable to new processing endpoints and system changes on existing endpoints, certification fees are charges for installation, upgrades, and cancellations. There is a minimum fee for a defined amount of certification time, though the client can schedule more time at an extra charge.
 - **File Maintenance Fees** – For Interlink and Visa check card issuers, these fees are charged for monthly residency, batch file updates, and account changes, additions, and deletions.
 - **Non-Financial Report and Raw Data Fees** – Fees assessed for all non-financial reports and raw data records.
 - **Processing Guarantees** – Processing guarantees paid to ensure a specified level of service. They are billed quarterly, based on overall volume listed in the client's operating certificate.
 - **Settlement/Reconciliation Charges** – Fees for additional funds transfers, after the daily complimentary transfer allowed each VisaNet endpoint.